

Exercise 2

The titanic dataset is a collection of data about 714 passengers, and the goal is to predict the survival (Survival: 1 if the passenger survived, 0 if they did not) based on some personal characteristics. In particular, here we consider the ticket class (Class: 1 = first, 0 = second or third; dummy), the gender (Gender: man = 1, woman = 0; dummy), and the age (Age, in years). Fitting a logistic regression model in R produces the following summary:

|                    | Estimate | Std. Error | z value | Pr(> z ) |
|--------------------|----------|------------|---------|----------|
| (Intercept)        | 1.5003   | 0.2462     | 6.09    | 0.0000   |
| Class              | 2.0103   | 0.2479     | 8.11    | 0.0000   |
| Gender             | -2.5473  | 0.2017     | ??      | 0.0000   |
| Age                | -0.0299  | 0.0074     | -4.06   | ??       |
|                    |          |            |         |          |
| Null deviance:     | 964.52   |            |         |          |
| Residual deviance: | 675.14   |            |         |          |

- a) Write the corresponding theoretical model and the expression of the estimated model.
- b) Write the likelihood and log-likelihood function.
- c) Complete the missing values in the table. For Pr(>|z|) of Age, write an approximate value. What variables are statistically significant?
- d) Provide an estimate of the odds for a woman aged 30 with a ticket of first class (denote this individual as “A”). How do you expect this value to change if you consider a person with the same characteristics but aged 31 (denote this individual as “B”)?
- e) Provide the interpretation of the coefficient associated with the Class variable. Given the estimate of this coefficient, what is the effect of this covariate on the survival probability?

n = 714

$$\log\left(\frac{\pi_i}{1-\pi_i}\right) = \eta_i = \begin{cases} 1 & \text{FIRST CLASS} \\ 0 & \text{OTHERWISE} \end{cases}$$
  
$$\eta_i = \beta_0 + \beta_1 D_1 + \beta_2 D_2 + \beta_3 x_3$$
  
$$Y_i \sim \text{BERNOULLI}(\pi_i)$$

- f) Perform a test  $H_0 : \beta_{\text{class}} = 0$  vs  $H_1 : \beta_{\text{class}} < 0$ .
- g) Define the “null deviance” and “residual deviance” in the output.
- h) Perform a test about the significance of the overall model.

6 LIKELIHOOD AND LOG-LIKELIHOOD

$Y_i \sim \text{BERNOULLI}(\pi_i) \quad f(\theta) = \pi_i^{y_i} (1-\pi_i)^{1-y_i} \quad \pi_i = \frac{e^{\tilde{x}_i \beta}}{1 + e^{\tilde{x}_i \beta}}$

LIKELIHOOD:

$$L(\theta) = \prod \pi_i^{y_i} (1-\pi_i)^{1-y_i}$$

LOG-LIKELIHOOD

$$\begin{aligned} l(\theta) = \log(L(\theta)) &= \log\left(\prod \pi_i^{y_i} (1-\pi_i)^{1-y_i}\right) = \sum \log(\pi_i^{y_i} (1-\pi_i)^{1-y_i}) = \sum y_i \log(\pi_i) + (1-y_i) \log(1-\pi_i) \\ &= \sum y_i \log\left(\frac{e^{\tilde{x}_i \beta}}{1 + e^{\tilde{x}_i \beta}}\right) + (1-y_i) \log\left(\frac{1}{1 + e^{\tilde{x}_i \beta}}\right) \\ &= \sum y_i \tilde{x}_i \beta - y_i \log(1 + e^{\tilde{x}_i \beta}) - \log(1 + e^{\tilde{x}_i \beta}) + y_i \log(1 + e^{\tilde{x}_i \beta}) = \sum y_i \tilde{x}_i \beta - \log(1 + e^{\tilde{x}_i \beta}) \end{aligned}$$

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|                       | Estimate | Std. Error | z value | Pr(> z ) |
|-----------------------|----------|------------|---------|----------|
| $\beta_0$ (Intercept) | 1.5003   | 0.2462     | 6.09    | 0.0000   |
| $\beta_1$ Class       | 2.0103   | 0.2479     | 8.11    | 0.0000   |
| $\beta_2$ Gender      | -2.5473  | 0.2017     | ??      | 0.0000   |
| $\beta_3$ Age         | -0.0299  | 0.0074     | -4.06   | ??       |
|                       |          |            |         |          |
| Null deviance:        | 964.52   |            |         |          |
| Residual deviance:    | 675.14   |            |         |          |

1  $\begin{cases} H_0: \beta_3 = 0 \\ H_1: \beta_3 \neq 0 \end{cases}$

$Z_{obs} \overset{H_0}{\sim} N(0,1) \quad Z_{obs} = \frac{\hat{\beta}_3 - \beta_3}{0.2017} = -12.62 \quad \alpha^{obs} = P(Z > |Z_{obs}|) = 2(1 - \Phi(|Z_{obs}|)) \approx 0$

• Since  $\alpha^{obs} < \alpha$  for  $\alpha = 0.1; \alpha = 0.05; \alpha = 0.01$  REJECT  $H_0$  AT ALL CONFIDENCE LEVEL  
• I PREFER THE MODEL WITH GENDER WRT THE MODEL WITHOUT GENDER

2  $\begin{cases} H_0: \beta_2 = 0 \\ H_1: \beta_2 \neq 0 \end{cases} \quad \alpha^{obs} = P(Z > |Z_{obs}|) = 2(1 - \Phi(|Z_{obs}|)) \approx 0$

$Z_{obs} = -9.06$

• Since  $\alpha^{obs} < \alpha$  for  $\alpha = 0.1; \alpha = 0.05; \alpha = 0.01$  REJECT  $H_0$  AT ALL CONFIDENCE LEVEL  
• I PREFER THE MODEL WITH AGE WRT THE MODEL W/O OUT AGE

THE VARIABLES THAT ARE STATISTICALLY SIGNIFICANT AT  $\alpha = 0.1; \alpha = 0.05$  AND  $\alpha = 0.01$  ARE THE ONE WITH  $\alpha^{obs} < \alpha$  SO ALL THE VARIABLES

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|              |              |
|--------------|--------------|
| INDIVIDUAL A | INDIVIDUAL B |
| • WOMAN      | • WOMAN      |
| • 30 YO      | • 31 YO      |
| • 1ST CLASS  | • 1ST CLASS  |

$$\frac{\pi_1}{\pi_0} = \frac{e^{\beta_0 + \beta_1 + \beta_2(x_0:1)}}{e^{\beta_0 + \beta_1 + \beta_2 x_0}} = e^{\beta_1} \rightarrow e^{\hat{\beta}_1} = e^{-0.0299} = 0.9705$$

THE LOG-ODDS OF SURVIVAL OF PATIENT B ARE EQUAL TO THE LOG-ODDS OF SURVIVAL OF PATIENT A MULTIPLIED BY 0.97

9 CLASS INTERPRETATION

WHEN KEEPING EVERYTHING CONSTANT, ON AVERAGE, IF WE MOVE TO 1ST CLASS WE INCREASE THE LOG ODDS OF SURVIVAL BY 2.0103

• 1ST CLASS HIGHER PROB. OF SURVIVING WRT 2ND AND 3RD CLASS

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$$\begin{cases} H_0: \beta_2 = 0 \\ H_1: \beta_2 < 0 \end{cases}$$

$$Z_2^{obs} \sim N(0,1) \quad Z_2^{obs} = \frac{\hat{\beta}_2 - \beta_2}{0.2479} = 3.11$$

$$d^{obs} = P(Z_2^{obs} | \alpha) \approx 1$$

NO X2 AND 1-SIDE  
1 TAIL

SINCE  $d^{obs} > \alpha$  FOR  $\alpha=0.1$ ;  $\alpha=0.05$ ;  $\alpha=0.01$   
DO NOT REJECT AT ALL CONFIDENCE LEVEL

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NULL DEVIANCE

$$D(NULL) = 2 \{ \tilde{l}(SAT.) - \tilde{l}(NULL) \}$$

- NULL MODEL  $\rightarrow$  NO COVARIATE
- SATURATED MODEL  $\rightarrow$  11 COVARIATES
- PARTIAL MODEL  $\rightarrow$  1 COVARIATE

RESIDUAL DEVIANCE

$$D(RES.) = 2 \{ \tilde{l}(SAT.) - \hat{l}(MODEL) \}$$

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$$\begin{cases} H_0: \beta_2 = \beta_3 = \beta_4 = 0 \\ H_1: 1HO \end{cases}$$

$$D(NULL) = 964.52$$

$$D(MOD) = 675.14$$

$$W \sim \chi^2_{4-1}$$

$$W = 2 \log \left( \frac{\hat{l}(MOD)}{\tilde{l}(NULL)} \right) = 2 \{ \hat{l}(MOD) - \tilde{l}(SAT.) + \tilde{l}(SAT.) - \tilde{l}(NULL) \}$$

$$= D(NULL) - D(MOD) = 964.52 - 675.14 = 289.38$$

$$d^{obs} = P(W > w_{obs}) = 1 - P(W \leq w_{obs}) \approx 0$$

SINCE  $d^{obs} < \alpha$  FOR  $\alpha=0.1$ ;  $\alpha=0.05$ ;  $\alpha=0.01$   
I REJECT NO I PREFER THE ORIGINAL  
MODEL WRT TO THE NULL MODEL